Abderrahmane FERRATI

Electrical and Industrial Control Systems Engineer

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Profile | Electrical engineering student with a strong interest in modeling, analysis, and control of industrial systems. Seeking a final-year project internship to deepen my knowledge of electrical engineering and control systems and apply it in an industrial environment.

Professional Experience

YAZAKI Morocco Meknes (YMM1) **Jun–Aug 2026**

Project Internship – Industrial Rooftop PV System Design

- Worked in a team to analyze electricity bills and roof constraints; selected components and ballasted mounting for a **1.20 MWp** system (**1,656 panels, 7 inverters**).
- Validated system sizing calculations in **PVsyst**: **2.18 GWh/year** and simulated demand coverage of **41% in summer**, **31% in winter**.
- Created 3D models and realistic renders in **Revit** to plan the rooftop layout.
- Developing a **demonstration dashboard**: instantaneous power, energy generation, and simulated faults.

ONCF – Rabat-Agdal **Jun–Aug 2025**

Introductory Internship – RGV Braking System Modernization under ERTMS

- Analyzed railway signaling equipment and identified faults in track circuits.
- Simulated P, PI, PD, and PID control strategies in MATLAB/Simulink to improve braking stability.

Education

Engineering Program – Electrical Engineering and Industrial Systems Control — ENSET Mohammedia — 2023 – Present

Preparatory Classes (MPSI/MP) — Lycée Omar Ibn Al Khattab — 2022 – 2024

Baccalaureate – Mathematical Sciences — Lycée Tarik — 2022

Academic Projects

Automated Filling and Capping Machine

- Designed an automated system controlled by a Siemens S7-300 PLC.
- Developed GRAFCET models for the process and control sections, designed industrial electrical wiring, and programmed LADDER logic in STEP 7.
- Ran integrated simulations in PLCSim and FluidSim, including normal, fault, and manual modes, and emergency stop.

24 V to 15 V Push-Pull Converter Using UC2525

- Designed a 50 kHz switch-mode converter using a high-frequency transformer, IRF3205 MOSFETs, and Schottky diodes.
- Simulated the circuit in Proteus (ISIS + ARES), designed the PCB, and implemented overcurrent and overtemperature protection (OCP, OTP) and soft-start.

Modular Training Platform – Electrical Engineering Test Bench Redesign

- Reverse-engineered and redesigned an existing three-phase transformer test bench.
- Replaced the three-phase transformer with three independent 100 VA multi-tap single-phase units, enabling single-phase and three-phase configurations.
- Redesigned the mechanical assembly in SolidWorks and implemented a universal six-diode rectifier.

Technical Skills

- **Renewable Energy** : PVsyst, Autodesk Revit, self-consumption PV system design, techno-economic analysis.
- **Industrial Automation** : Siemens TIA Portal, STEP 7, GRAFCET, LADDER, FluidSim, S7-300/S7-1500 PLCs.
- **Electrical Engineering** : CANECO BT, AutoCAD Electrical, power and control schematics, cable sizing.
- **Embedded Systems** : Arduino, C, sensors, data acquisition, human-machine interfaces.
- **Simulation & Modeling** : MATLAB/Simulink, Proteus.
- **Programming** : Python, C, LADDER, Structured Text (ST).
- **Standards & Safety** : IEC 60364, IEC 61131-3, NF C 15-100, ISO 50001.

Languages

- Arabic (native), French (fluent), English (C1 – EF SET)

Certifications

– Python Essentials 1 & 2, Cisco Networking Academy

– ISO 50001:2018 Awareness

– EF SET English Certificate, C1